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PAPER_1105: Hydrogen-Universe Dual 3D MUGE: Scale Invariance from Bohr Radius to Hubble Horizon

Abstract

We construct a dual 3D Modified Universal Gravity Equation (MUGE) system comparing the hydrogen atom (quantum scale, $r = a_0$) to the observable universe (cosmological scale, $r = R_H$), demonstrating UQFF scale invariance through the 9-term compressed MUGE framework evaluated on volumetric grids at both scales.

1. The 9-Term Compressed MUGE

$$g_{\text{MUGE}}(r) = g_N + g_{\text{exp}} + g_{\text{super}} + g_{\text{env}} + g_{\text{Ug}} + g_{\text{cosm}} + g_Q + g_{\text{fluid}} + g_{\text{pert}}$$

Term Definitions

#	Term	Expression	Physics
1	g_N	GM/r^2	Newtonian gravity
2	g_{exp}	$H_0^2\, r$	Hubble expansion
3	g_{super}	$B^2/(2\mu_0 \rho_{\text{avg}}\, r)$	Magnetic suppression
4	g_{env}	$\kappa\, g_N\, \exp(-1/100)$	Envelope dissipation
5	g_{Ug}	$\sum_{i=1}^{26} Q_i\, g_N/26$	26-layer Ug aggregate
6	g_{cosm}	$(\Lambda c^2/3)\, r$	Cosmological constant
7	g_Q	$\hbar/(M r^3)$	Quantum correction
8	g_{fluid}	ν/r^2	Navier-Stokes viscous
9	g_{pert}	$(4\pi/3) G \rho_{\text{DM}}\, r$	Dark matter perturbation

2. Hydrogen Scale ($r = a_0$)

Parameters: $M = m_p = 1.673 \times 10^{-27}$ kg, $r = a_0 = 5.292 \times 10^{-11}$ m.

Dominant Terms

$$g_N^{(H)} = \frac{G m_p}{a_0^2} \approx 3.99 \times 10^{-8}\ \text{m/s}^2$$

$$g_Q^{(H)} = \frac{\hbar}{m_p\, a_0^3} \approx 4.25 \times 10^{24}\ \text{m/s}^2$$

The quantum correction dominates by $\sim$ 32 orders of magnitude at the Bohr radius, confirming that atomic-scale MUGE is quantum-dominated.

3. Universe Scale ($r = R_H$)

Parameters: $M = M_U = 1.5 \times 10^{53}$ kg, $r = R_H = 4.4 \times 10^{26}$ m.

Dominant Terms

$$g_N^{(U)} = \frac{GM_U}{R_H^2} \approx 5.17 \times 10^{-5} \text{ m/s}^2$$

$$g_{\text{cosm}}^{(U)} = \frac{\Lambda c^2}{3} R_H \approx 1.44 \times 10^{-5} \text{ m/s}^2$$

At the Hubble radius, Newtonian and cosmological-constant terms are comparable, with the Hubble expansion term $g_{\text{exp}}^{(U)} = H_0^2 R_H \approx 2.13 \times 10^{-9}$ m/s^2 subdominant.

4. Scale Invariance Ratio

$$\mathcal{R} = \frac{g_{\text{MUGE}}^{(H)}}{g_{\text{MUGE}}^{(U)}}$$

This ratio encodes the full UQFF scale-invariance structure. The fact that both systems are described by the same 9-term equation with only mass and radius as inputs demonstrates MUGE universality.

5. 3D Volumetric Grid

A $5^3 = 125$-point spatial grid is constructed at each scale, evaluating g_{MUGE} at each point $\mathbf{r}_{\text{grid}}$ centered on the system. Grid statistics (average, standard deviation) characterize the spatial profile of the gravitational field at each scale.

6. Implementation

Calculator: `HydrogenUniverseDual3DMUGECalculator` in CondensedPhysics.py

- Default 5^3 grid (configurable via `n_grid`)
- Outputs: single-point and grid-averaged MUGE for both scales, scale ratio, per-term breakdown

7. Conclusion

The dual hydrogen-universe 3D MUGE system demonstrates that the UQFF 9-term framework spans 37 orders of magnitude in length scale (10^{-11} to 10^{26} m), transitioning smoothly from quantum-dominated physics to cosmological-constant-dominated physics through a single unified equation set.

References

- Murphy, D.T. (2025). UQFF: Star Cradle Mechanics framework. Star-Magic repository.
- PAPER_491: MUGE Compressed Nine-Term Gravity Framework.